Building Materials

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Chapter One: Introduction

Why we offer this course?

Why do you take this course?

Chapter One: Introduction

Sub-disciplines of civil engineering:

- Environmental engineering
- Geotechnical engineering
- Structural engineering
- Transportation engineering
- Water resources engineering
- Surveying
- Urban planning
- Construction engineering
- Materials engineering

Chapter One: Introduction

1. **Materials and Structures**
2. **Sustainable Development of Building Materials**
3. **Components – Structure – Performance**
4. **Syllabus of This Course**

Chapter One: Introduction

Building material

- Any material which is used for construction purpose is called as building material.

- Natural materials
  - Unprocessed, raw:
  - Minimally processed

  Wood, bamboo, soil, metal …
- Synthetic materials
  - Industrial processing

  Cement, plastic, composites….

1.1 Materials and Engineering Structures

----- Wood

Advantage:

Good insulation, strength， safe properties…

weakness:

Bad fire resistance…

1.1 Materials and Engineering Structures

----- Stone

Advantage: 

High strength, good fire resistance and durability…

weakness: 

Bad insulation…

1.1 Materials and Engineering Structures

----- Masonry

Advantage:

Good fire protection, a useful life of more than 500 …

weakness:

much lower in tensile strength…

1.1 Materials and Engineering Structures

----- Iron and Steel

Advantage:

Strong, stable …

weakness:

Heat conductivity, corrosion…

1.1 Materials and Engineering Structures

----- Asphalt

Advantage:

Recyclable , economical, durable …

weakness:

Poor fire resistance, close to depletion…

What is the most widely used building materials in the world?

1.1 Materials and Engineering Structures

----- Concrete

人造“岩石”（砼）。

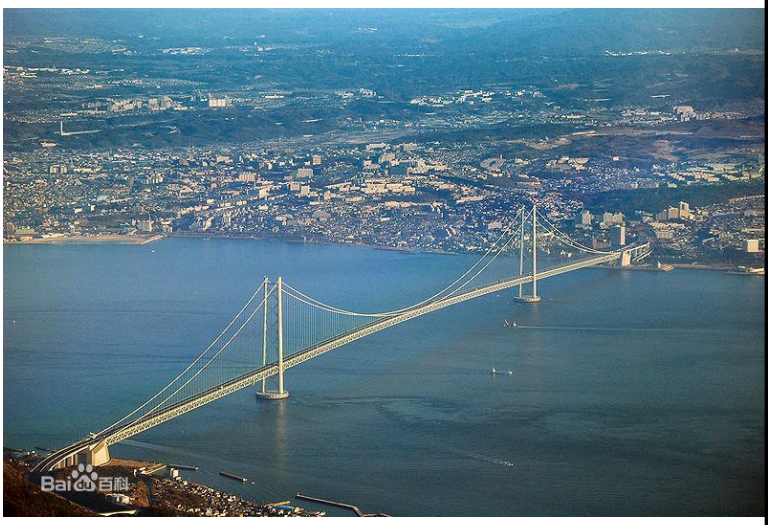




1.1 Materials and Engineering Structures

----- Concrete

The Tatara bridge in Japan is the world's longest cable-stayed concrete bridge.



1.1 Materials and Engineering Structures

----- Concrete

The Burj skyscraper in Dubai is the world's tallest building (828m high).



1.1 Materials and Engineering Structures



1.1 Materials and Engineering Structures

Admixtures make the high-rise building, and steel-concrete structure construction possible





Chapter One: Introduction

1. Materials and Engineering Structures
2. Sustainable Development of Building Materials
3. Components – Structure – Performance
4. Course Syllabus

1.2 Sustainable Development of Building Materials

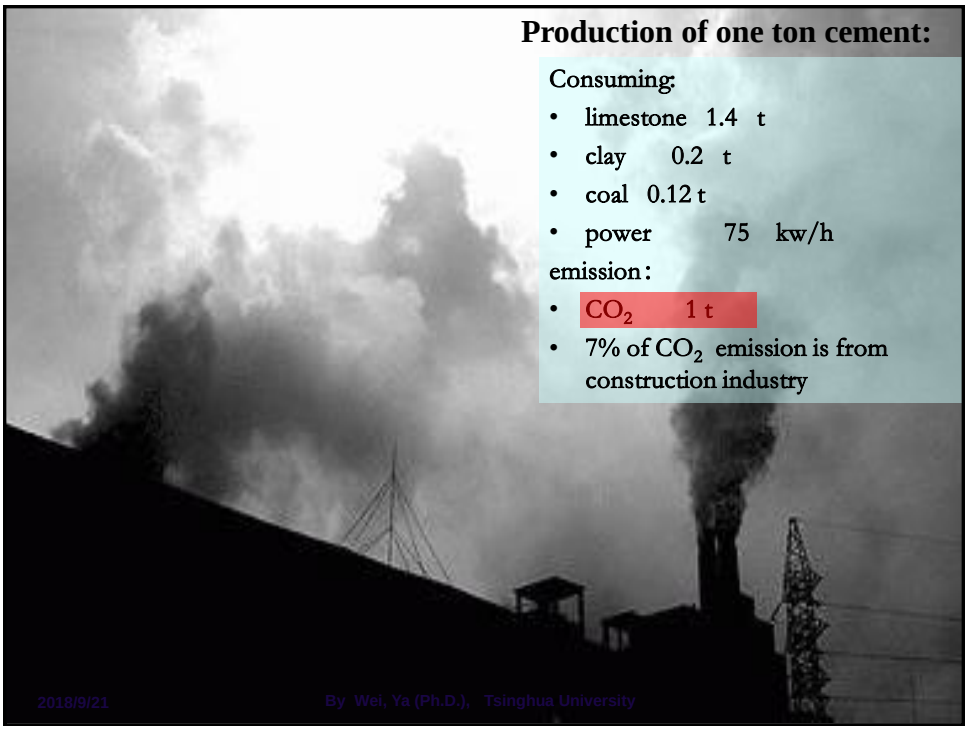
Problem from concrete industry:

- CO_2 emission
- Consumption of energy and resource
- wastes









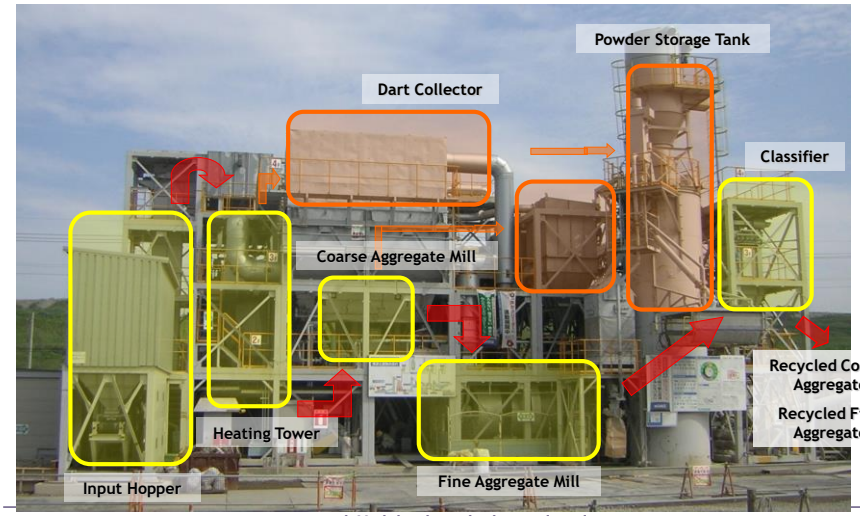
1.2 Sustainable Development of Building Materials

Cement production in China

Year	Production （亿吨）
1980	0.8
1985	1.43 (US/0.88, JPN/0.95, USSR/1.4)
1997	> 5
2000	~ 6
2001	6.4
2002	7
2003	7.5 (人均 ~750 Kg)
…	
2006	10.6 (人均 ~820 kg) / 全球产量：22.7
2017	23.2 全球产量：47.4

1.2 Sustainable Development of Building Materials

Recycling old concrete:



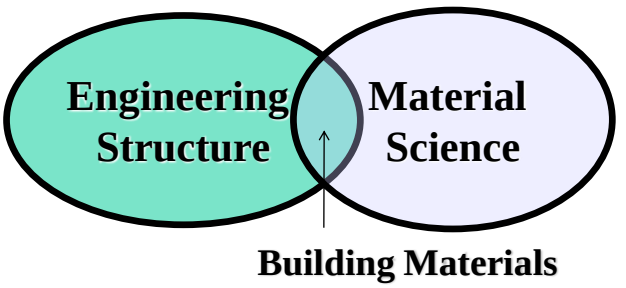


A four-lane pavement about 100km long in Guangdong Province.
20~35% of fly ash added in the pavement.

Chapter One: Introduction

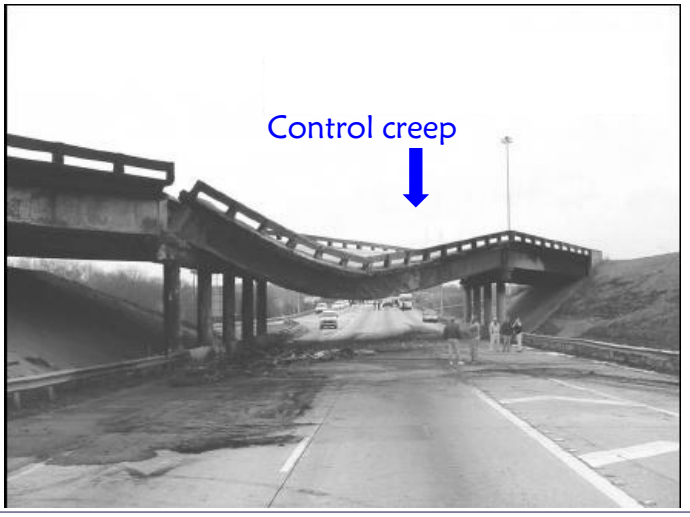
1. Materials and Engineering Structures
2. Sustainable Development of Building Materials
3. Components – Structure – Performance
4. Course Syllabus

1.3 Components -- Structure -- Performance



1.3 Components -- Structure -- Performance

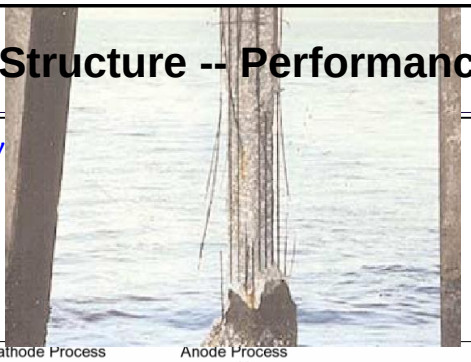
What cause the damage?

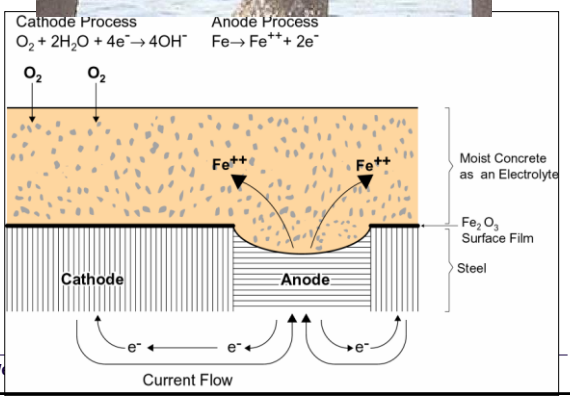


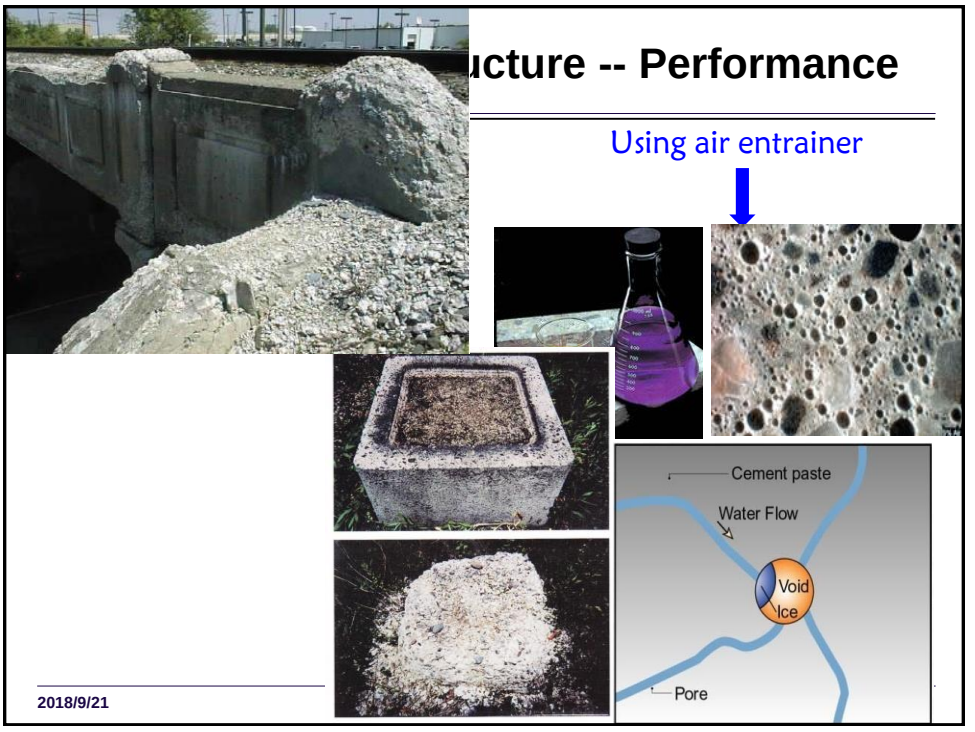
1.3 Components -- Structure -- Performance

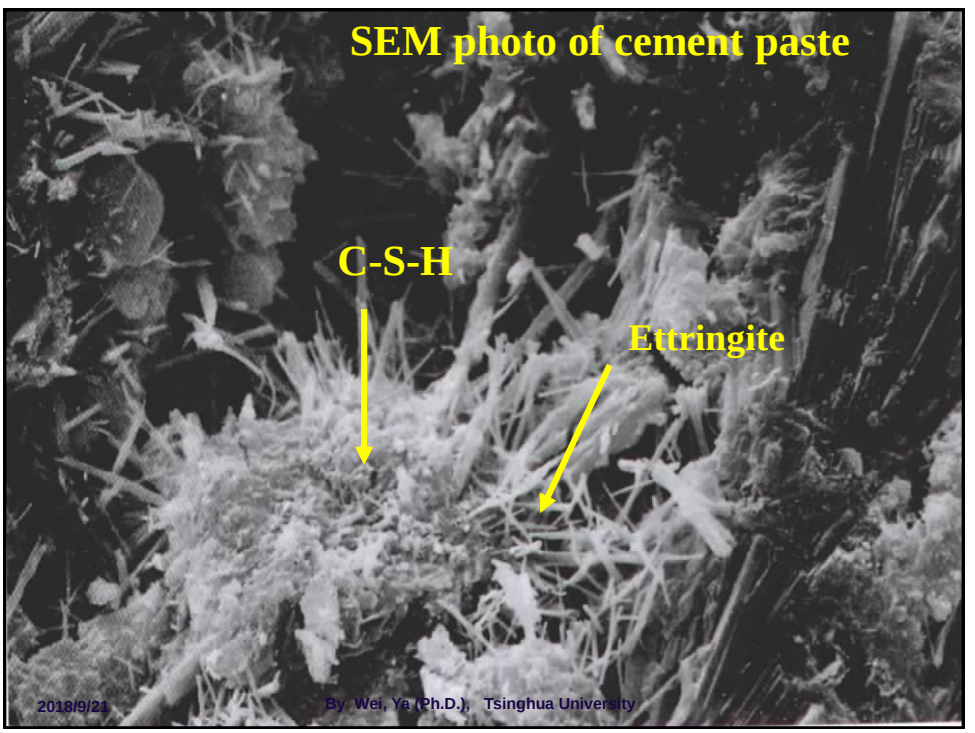
Control permeability





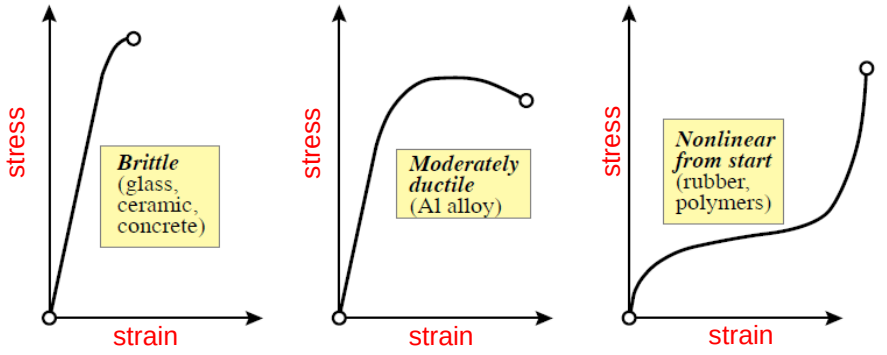



ucture -- Performance
Using air entrainer
Cement paste
Water Flow
Void
Ice
Pore
2018/9/21


SEM photo of cement paste
C-S-H
Ettringite
2018/9/21
By Wei, Ya (Ph.D.), Tsinghua University

1.3 Components -- Structure -- Performance

Mechanical Properties



1.3 Components -- Structure -- Performance

Nonmechanical Properties





1.3 Components -- Structure -- Performance

Nonmechanical Properties

Thermal expansion:

A temperature change of △T with respect to a base or reference level produces a thermal strain :

$$\varepsilon_T = \alpha \cdot \Delta T$$

Where α is the coefficient of thermal dilation, measured in 1/℉ or 1/℃.

Material	$\alpha \times 10^{-6}$
Plastics	~ 100
Aluminum	24
Copper	20
Steel	15
G-10 Fiberglass	9
Wood	5
Glass	3–5
Invar	1.5
Fused Silica	0.6

1.3 Components -- Structure -- Performance

Nonmechanical Properties

Thermal expansion:

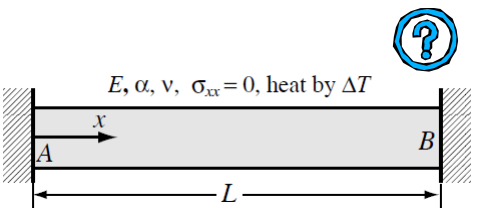
Example:

Calculate the change in length of a 120m steel pipe that will be expected to temperatures ranging from 12 to 95 ℃.

$$\delta = \alpha L(\Delta T) = 0.0000117 \times 120 \times (95 - 12) = 0.12m$$

To combine mechanical and thermal effects:

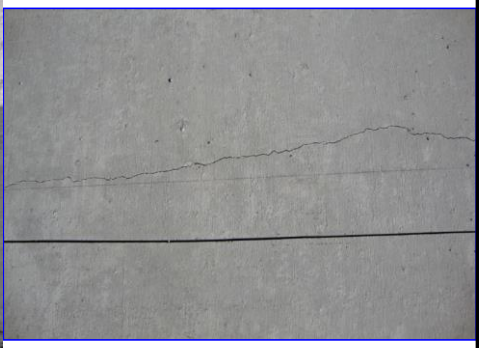
$$\varepsilon = \varepsilon_M + \varepsilon_T = \frac{\sigma}{E} + \alpha \cdot \Delta T$$



1.3 Components -- Structure -- Performance

Nonmechanical Properties





1.3 Components -- Structure -- Performance

Laboratory measuring devices

Linear Variable Differential Transformer (LVDT)

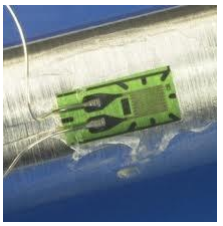


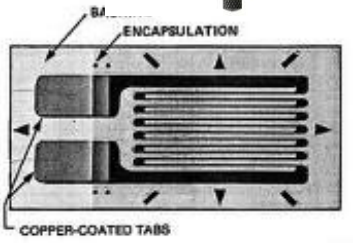
Dial gauge



Strain gauge







1.3 Components -- Structure -- Performance

Laboratory measuring devices

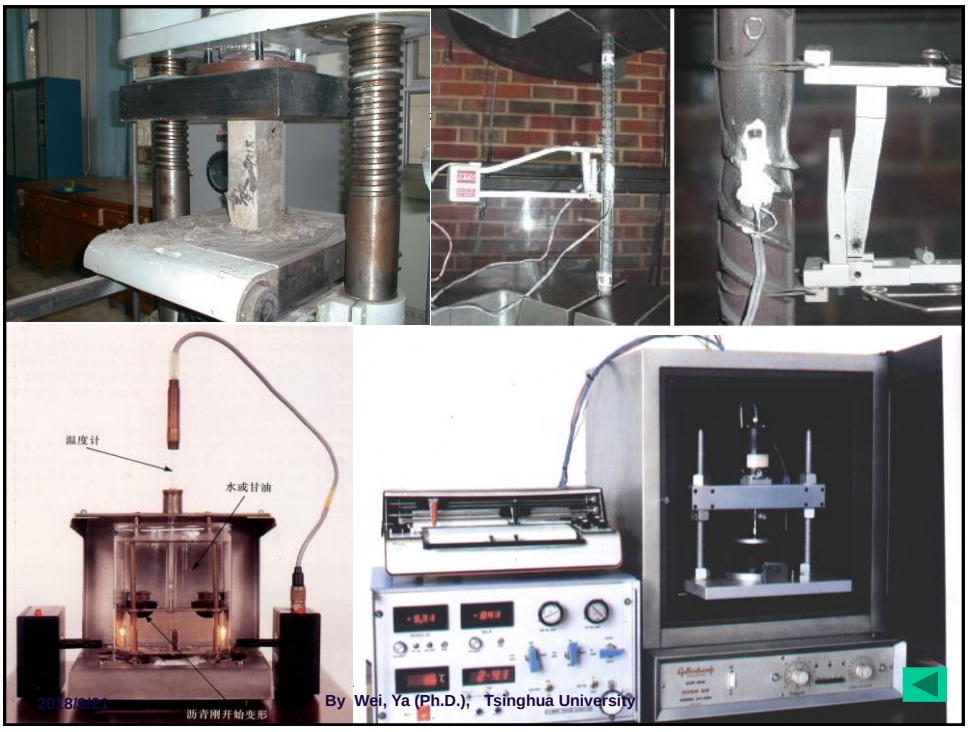
Proving ring



Load cell







Chapter One: Introduction

1. **Materials and Engineering Structures**
2. **Sustainable Development of Building Materials**
3. **Components – Structure – Performance**
4. **Course Syllabus**

Chapter One	Introduction (1)
Chapter Two	The Fundamentals (2)
Chapter Three	Metal (2)
Chapter Four	Concrete (6)
Term project presentation (2)	
Chapter Five	Asphalt (1.5)
Chapter Six	Masonry and Wood (0.5)

1.4 Course Syllabus

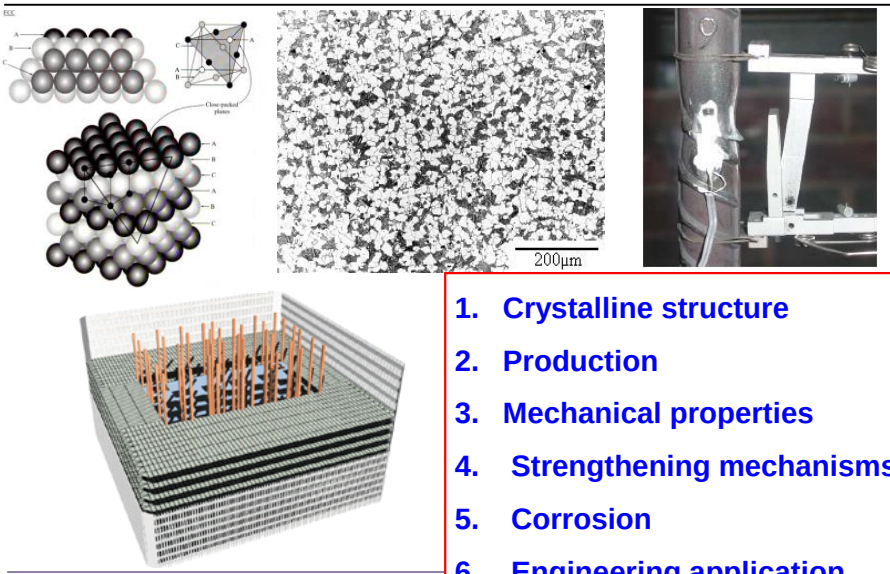
-- Fundamentals

- **How do they behave in service?**
  e.g. strength, strain, viscosity
- **Why do they behave in the way that they do?**
  composition, bonding, structure, surface properties

1. **Physical Architecture of Materials**
2. **Behavior of Materials Under Stress**
3. **Surface Properties**
4. **Failure and Fracture**

1.4 Course Syllabus

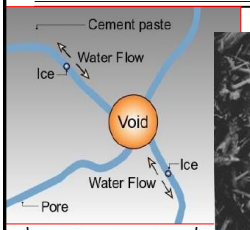
-- Metal

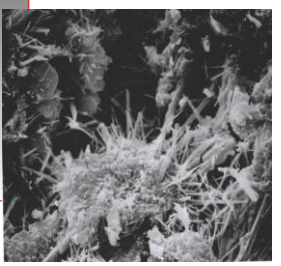


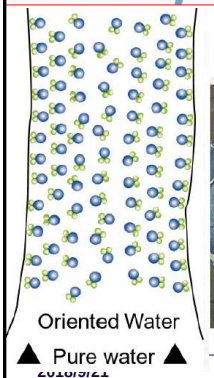
1. **Crystalline structure**
2. **Production**
3. **Mechanical properties**
4. **Strengthening mechanisms**
5. **Corrosion**
6. **Engineering application**

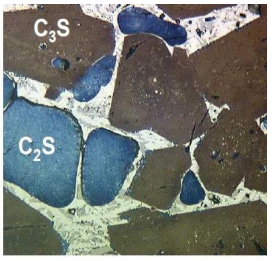
1.4 Course Syllabus

-- Concrete





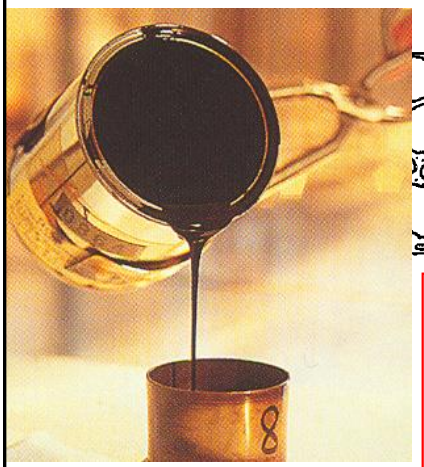


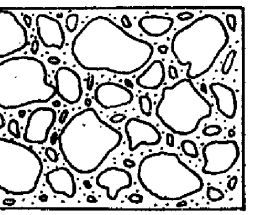


1. Compositions
2. Structure
3. Strength and Damage
4. Mix Proporationing
5. Early-age Properties
6. Volume Changes
7. Durability

1.4 Course Syllabus

-- Asphalt



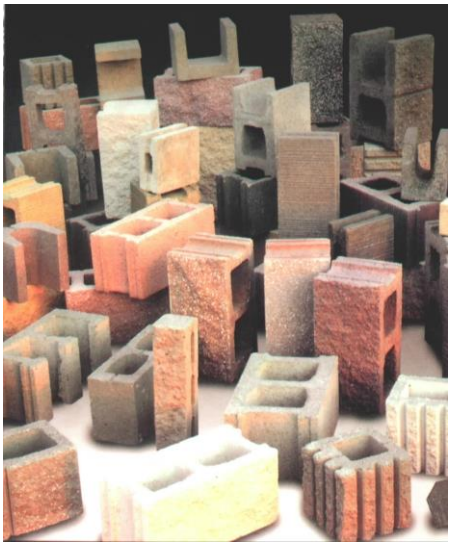


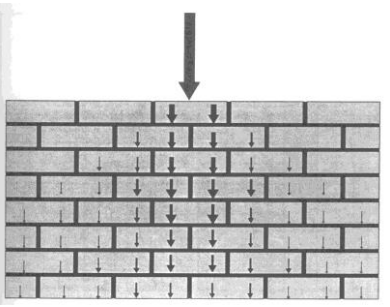


1. Characterization of Asphalt
2. Asphalt Binder
3. Asphalt Concrete Mix Design
4. Engineering Applications

1.4 Course Syllabus

-- Masonry

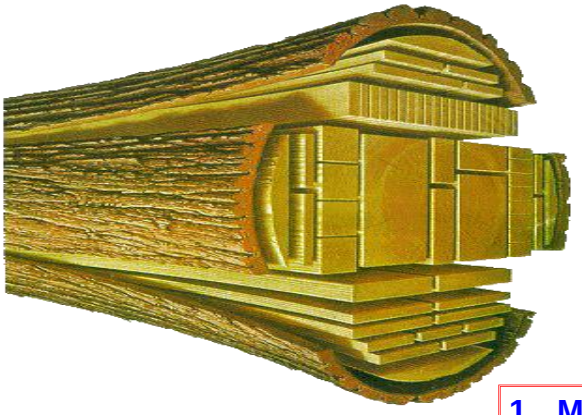




1. Introduction
2. Masonry Unit
3. Mortar, Grout and Plaster
4. Properties of Masonry

1.4 Course Syllabus

-- Wood



1. Moisture Relations
2. Physical Property
3. Mechanical Property

1.4 Course Syllabus

-- Expectations

1. Attendance is expected all class.
2. Each assignment submitted must represent essentially your own work.
3. Homework assignments should be neat, complete, and logically arranged.

1.4 Course Syllabus

-- Textbooks and References

1. Concrete: Structure, Properties and Materials, by Mehta P.K. et al., Prentice Hall.
2. 建筑材料，张君，阎培渝，覃维祖 编著，清华大学出版社，2007

1.4 Course Syllabus

-- Teaching Assistant

Wu，Zehong（吴泽弘）
Graduate Student

wzhthu@163.com

1.4 Course Syllabus

-- Grading Policy

Course Grading:	**% of Total Grade**
Quizzes	10
Homework Assignments (~ 13)	15
Term Report (1)	15
Final Exam (Open books and notes)	60
Total	**100**

Missing hws is not allowed in this class, otherwise, one will have no right to take the final exam….

作业缺交2次及以上的同学不能参加期末考试。



QUESTIONS?